

FINANCIAL MANAGEMENT I (ACC 413)

LECTURE NOTES

INTRODUCTION:

One needs money to make money. Finance is the life-blood of business and there must be a continuous flow of funds in and out of a business enterprise. Money makes the wheels of business run smoothly. Sound plans, efficient production system and excellent marketing network are all hampered in the absence of an adequate and timely supply of funds.

Sound financial management is as important in business as production and marketing. A business firm requires finance to commence its operations, to continue operations and for expansion or growth. Finance is, therefore, an important operative function of business.

A large business firm has to raise funds from several sources and has to utilise those funds in alternative investment opportunities. In order to ensure the most judicious utilisation of funds and to provide a reasonable rate of return on the investment, sound financial policies and programmes are required. Unwise financing can drive a business into bankruptcy just as easily as a poor product, inept marketing or high production costs.

On the other hand, adequate and economical financing can provide the firm a differential advantage in the market place. The success of a business enterprise is largely determined by the way its capital funds are raised, utilised and disbursed. In the modern money-using economy, the importance of finance has increased further due to increasing scale of operations and capital-intensive techniques of production and distribution.

In fact, finance is the bright thread running through all business activity. It influences and limits the activities of marketing, production, purchasing and personnel management. The success of a business is measured largely in financial terms. The efficient organisation and administration of the finance function is thus vital to the successful functioning of every business enterprise.

Meaning of Financial Management:

Financial Management means planning, organizing, directing and controlling the financial activities such as procurement and utilization of funds of the enterprise. It means applying general management principles to financial resources of the enterprise.

Financial management may be defined as planning, organising, directing and controlling the financial activities of an organisation.

According to Guthman and Dougal, financial management means, “the activity concerned with the planning, raising, controlling and administering of funds used in the business.” It is concerned with the procurement and utilisation of funds in the proper manner.

“Financial management is that area of business management devoted to a judicious use of capital and a careful selection of the source of capital in order to enable a spending unit to move in the direction of reaching the goals.”

– **J.F. Brandley**

“Financial management is the operational activity of a business that is responsible for obtaining and effectively utilizing the funds necessary for efficient operations.” - **Massie**

Financial activities deal with not only the procurement and utilisation of funds but also with the assessing of needs for funds, raising required finance, capital budgeting, distribution of surplus, financial controls, etc.

Ezra Solomon has described the nature of financial management as follows:

“Financial management is properly viewed as an integral part of overall management rather than as a staff specially concerned with funds raising operations.

In this broader view, the central issue of financial policy is the wise use of funds and the central process involved is a rational matching of the advantage of potential uses against the cost of alternative potential sources so as to achieve the broad financial goals which an enterprise set for itself.

In addition to raising funds, financial management is directly concerned with production, marketing and other functions within an enterprise whenever decisions are made about the acquisition or distribution of funds.”

Nature, Significance, and Scope of Financial Management

The introduction to financial management also requires you to understand the scope of financial management. It is important that financial decisions take care of the shareholders interests.

Further, they are upheld by the maximization of the wealth of the shareholders, which depends on the increase in net worth, capital invested in the business, and ploughed-back profits for the growth and prosperity of the organization.

The scope of financial management is explained in the diagram below:

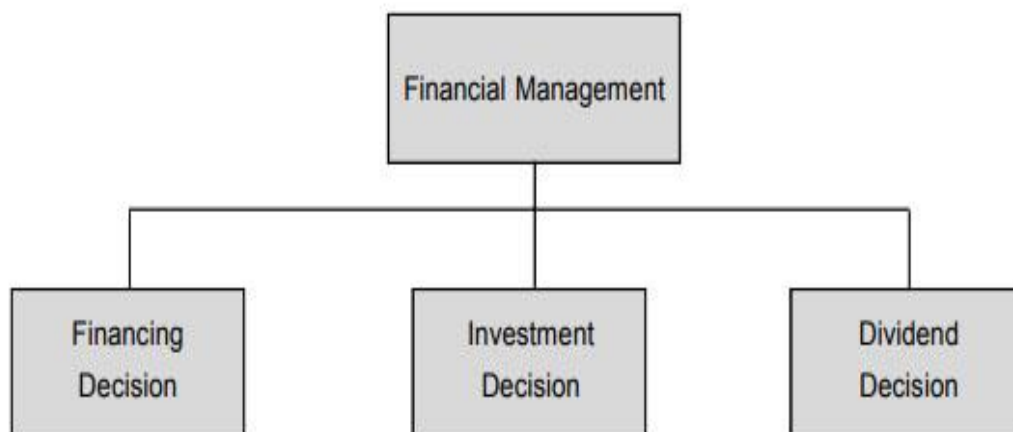


Fig. 1 - The scope of Financial Management

You can understand the nature of financial management by studying the nature of investment, financing, and dividend decisions.

Core Financial Management Decisions

In organizations, managers in an effort to minimize the costs of procuring finance and using it in the most profitable manner, take the following decisions:

Investment Decisions: Managers need to decide on the amount of investment available out of the existing finance, on a long-term and short-term basis. They are of two types:

- Long-term investment decisions or Capital Budgeting mean committing funds for a long period of time like fixed assets. These decisions are irreversible and usually include the ones pertaining to investing in a building and/or land, acquiring new plants/machinery or replacing the old ones, etc. These decisions determine the financial pursuits and performance of a business.
- Short-term investment decisions or Working Capital Management means committing funds for a short period of time like current assets. These involve decisions pertaining to the investment of funds in the inventory, cash, bank deposits, and other short-term investments. They directly affect the liquidity and performance of the business.

Financing Decisions: Managers also make decisions pertaining to raising finance from long-term sources (called Capital Structure) and short-term sources (called Working Capital). They are of two types:

- **Financial Planning decisions** which relate to estimating the sources and application of funds. It means pre-estimating financial needs of an organization to ensure the availability of adequate finance. The primary objective of financial planning is to plan and ensure that the funds are available as and when required.
- **Capital Structure decisions** which involve identifying sources of funds. They also involve decisions with respect to choosing external sources like issuing shares, bonds, borrowing from banks or internal sources like retained earnings for raising funds.

Dividend Decisions: These involve decisions related to the portion of profits that will be distributed as dividend. Shareholders always demand a higher dividend, while the management would want to retain profits for business needs. Hence, this is a complex managerial decision.

Financial management is an organic function of any business. Any organization needs finances to obtain physical resources, carry out the production activities and other business operations, pay compensation to the suppliers, etc. There are many theories around financial management:

1. Some experts believe that financial management is all about providing funds needed by a business on terms that are most favourable, keeping its objectives in mind. Therefore, this approach concerns primarily with the procurement of funds which may include instruments, institutions, and practices to raise funds. It also takes care of the legal and accounting relationship between an enterprise and its source of funds.
2. Another set of experts believe that finance is all about cash. Since all business transactions involve cash, directly or indirectly, finance is concerned with everything done by the business.
3. The third and more widely accepted point of view is that financial management includes the procurement of funds and their effective utilization. For example, in the case of a manufacturing company, financial management must ensure that funds are available for installing the production plant and machinery. Further, it must also ensure that the profits adequately compensate the costs and risks borne by the business.

In a developed market, most businesses can raise capital easily. However, the real problem is the efficient utilization of the capital through effective financial planning and control.

Further, the business must ensure that it deals with tasks like ensuring the availability of funds, allocating them, managing them, investing them, controlling costs, forecasting financial requirements, planning profits and estimating returns on investment, assessing working capital, etc.

Objectives of Financial Management

The financial management is generally concerned with procurement, allocation and control of financial resources of a concern. The objectives can be-

1. To ensure regular and adequate supply of funds to the concern.
2. To ensure adequate returns to the shareholders which will depend upon the earning capacity, market price of the share, expectations of the shareholders.

3. To ensure optimum funds utilization. Once the funds are procured, they should be utilized in maximum possible way at least cost.
4. To ensure safety on investment, i.e, funds should be invested in safe ventures so that adequate rate of return can be achieved.
5. To plan a sound capital structure-There should be sound and fair composition of capital so that a balance is maintained between debt and equity capital.

Objectives of Financial Management:

Financial management is one of the functional areas of business. Therefore, its objectives must be consistent with the overall objectives of business. The overall objective of financial management is to provide **maximum return** to the owners on their investment in the long- term.

This is known as **wealth maximisation**. Maximisation of owners' wealth is possible when the capital invested initially increases over a period of time. Wealth maximisation means maximising the market value of investment in shares of the company.

Wealth of shareholders = Number of shares held × Market price per share.

In order to maximise wealth, financial management must achieve the following specific objectives:

- (a) To ensure availability of sufficient funds at reasonable cost (liquidity).
- (b) To ensure effective utilisation of funds (financial control).
- (c) To ensure safety of funds by creating reserves, re-investing profits, etc. (minimisation of risk).
- (d) To ensure adequate return on investment (profitability).
- (e) To generate and build-up surplus for expansion and growth (growth).
- (f) To minimise cost of capital by developing a sound and economical combination of corporate securities (economy).
- (g) To coordinate the activities of the finance department with the activities of other departments of the firm (cooperation).

Profit Maximisation:

Very often maximisation of profits is considered to be the main objective of financial management. Profitability is an operational concept that signifies economic efficiency. Some writers on finance believe that it leads to efficient allocation of resources and optimum use of capital.

It is said that profit maximisation is a simple and straightforward objective. It also ensures the survival and growth of a business firm. But modern authors on financial management have criticised the goal of profit maximisation.

Ezra Solomon has raised the following objections against the profit maximisation objective:

Objections against the Profit Maximisation Objectives:

- (i) The concept is ambiguous or vague. It is amenable to different interpretations, e.g., long run profits, short run profits, volume of profits, rate of profit, etc.

- (ii) It ignores the timing of returns. It is based on the assumption of bigger the better and does not take into account the time value of money. The value of benefits received today and those received a year later are not the same.
- (iii) It ignores the quality of the expected benefits or the risk involved in prospective earnings stream. The streams of benefits may have varying degrees of uncertainty. Two projects may have same total expected earnings but if the earnings of one fluctuate less widely than those of the other it will be less risky and more preferable. More uncertain or fluctuating the expected earnings, lower is their quality.
- (iv) It does not consider the effect of dividend policy on the market price of the share. The goal of profit maximisation implies maximising earnings per share which is not necessarily the same as maximising market-price share. According to Solomon, “to the extent payment of dividends can affect the market price of “the stock (or share), the maximisation of earnings per share will not be a satisfactory objective by itself.”
- (v) Profit maximisation objective does not take into consideration the social responsibilities of business. It ignores the interests of workers, consumers, government and the public in general. The exclusive attention on profit maximisation may misguide managers to the point where they may endanger the survival of the firm by ignoring research, executive development and other intangible investments.

Wealth Maximisation:

Prof. Ezra Solomon has advocated wealth maximisation as the goal of financial decision-making. Wealth maximisation or net present worth maximisation is defined as follows: “The gross present worth of a course of action is equal to the capitalised value of the flow of future expected benefits, discounted (or as capitalised) at a rate which reflects their certainty or uncertainty.

Wealth or net present worth is the difference between gross present worth and the amount of capital investment required to achieve the benefits being discussed. Any financial action which creates wealth or which has a net present worth above zero is a desirable one and should be undertaken.

Any financial action which does not meet this test should be rejected. If two or more desirable courses of action are mutually exclusive (i.e., if only one can be undertaken), then the decision should be to do that which creates most wealth or shows the greatest amount of net present worth. In short, the operating objective for financial management is to maximise wealth or net present worth.”

Wealth maximisation is more operationally viable and valid criterion because of the following reasons:

- (a) It is a precise and unambiguous concept. The wealth maximisation means maximising the market value of shares.
- (b) It takes into account both the quantity and quality of the expected stream of future benefits. Adjustments are made for risk (uncertainty of expected returns) and timing (time value of money) by discounting the cash flows,
- (c) As a decision criterion, wealth maximisation involves a comparison of value of cost. It is a long-term strategy emphasising the use of resources to yield economic values higher than joint values of inputs.

(d) Wealth maximisation is not in conflict with the other motives like maximisation of sales or market share. It rather helps in the achievement of these other objectives. In fact, achievement of wealth maximisation also maximises the achievement of the other objectives. Therefore, maximisation of wealth is the operating objective by which financial decisions should be guided.

The above description reveals that wealth maximisation is more useful if objective than profit maximisation. It views profits from the long-term perspective. The true index of the value of a firm is the market price of its shares as it reflects the influence of all such factors as earnings per share, timing of earnings, risk involved, etc.

Thus, the wealth maximisation objective implies that the objective of financial management should be to maximise the market price of the company's shares in the long-term. It is a true indicator of the company's progress and the shareholder's wealth.

However, "profit maximisation can be part of a wealth maximisation strategy. Quite often the two objectives can be pursued simultaneously but the maximisation of profits should never be permitted to overshadow the broader objectives of wealth maximisation.

Functions of Financial Management

1. **Estimation of capital requirements:** A finance manager has to make estimation with regards to capital requirements of the company. This will depend upon expected costs and profits and future programmes and policies of a concern. Estimations have to be made in an adequate manner which increases earning capacity of enterprise.
2. **Determination of capital composition:** Once the estimation has been made, the capital structure has to be decided. This involves short- term and long- term debt equity analysis. This will depend upon the proportion of equity capital a company is possessing and additional funds which have to be raised from outside parties.
3. **Choice of sources of funds:** For additional funds to be procured, a company has many choices like-
 - a. Issue of shares and debentures
 - b. Loans to be taken from banks and financial institutions
 - c. Public deposits to be drawn like in form of bonds.

Choice of factor will depend on relative merits and demerits of each source and period of financing.

4. **Investment of funds:** The finance manager has to decide to allocate funds into profitable ventures so that there is safety on investment and regular returns is possible.
5. **Disposal of surplus:** The net profits decision has to be made by the finance manager. This can be done in two ways:
 - a. Dividend declaration - It includes identifying the rate of dividends and other benefits like bonus.
 - b. Retained profits - The volume has to be decided which will depend upon expansional, innovation, diversification plans of the company.
6. **Management of cash:** Finance manager has to make decisions with regards to cash management. Cash is required for many purposes like payment of wages and salaries, payment of electricity and water bills, payment to creditors, meeting current liabilities, maintenance of enough stock, purchase of raw materials, etc.
7. **Financial controls:** The finance manager has not only to plan, procure and utilize the funds but he also has to exercise control over finances. This can be done through many techniques like ratio analysis, financial forecasting, cost and profit control, etc.

Importance of Financial Management in an Organization

Finance is actually that soul out of business relationship. Importance of financial management is to meet each prerequisite of the business concern. Every business concern should keep up sufficient finance with regards to their smooth functioning of their business operations. Also, in addition continue growth is one of the importance of financial management in an organization to achieve the future goal. Now days, there is necessity to give importance of financial management in modern business to achieve objective related to finance of a company. People cannot neglect the importance of financial management in business planning and controlling at all conditions or stages of your business.

Top 10 – Importance of Financial Management Points:

There is a huge importance of financial management in business planning and controlling for your financial stability and to keep you away from bankruptcy. Here we will see what is the importance of financial management in points mentioned below:

1. Financial Planning:

Financial management its importance is financial planning. It decides each financial necessity associated with business concern. Also, financial planning associates need to take prompts and correct measures instead of worries in later stage of financial management life-cycle of a company. Financial planning looks a crucial area associated with business concern. Typically, all the credit for business success is mostly depends on the financial planning of a company.

2. Safeguarding / Protecting Funds:

Importance of financial management include protecting finance towards achieving business goals. One has to measure the areas where funds are required and allocate it well in all the areas for smooth functioning of business. Overspending on one project and impact other business operations as they may lack finance in many cases. It is crucial to safeguard funds and invest wisely.

3. Allocation of Funds:

Importance of financial management in an organization is to allocate funds appropriately. When making proper use of allocated finance to assets enhance the operational proficiency for the business concern. Whenever the finance specialists make use of the funds appropriately and allocate it wisely, they can reduce business expense and increase capital estimated for a company.

4. Investment Opportunities:

As a person, if you are good at managing your finance and saving then you get opportunities to explorer investment. Investment opportunities will assist you in creating wealth so that you can enjoy your retirement period. There are various investment opportunities you can explorer like investing in stocks, gold, mutual funds, property, lands, etc. You can study about investing in detail to know the

risk and return of investment. Depending upon your risk ability you can then choose the appropriate investment options.

5. Financial Decision:

Its importance of financial management points is financial decision. Once financial choice according to the business concern has made, it cannot be rewind. As finance once spend will not be repaid again for any wrong decision made. Financial selection might impact the whole business operation. Since it has an instant relationship with all the departments of a company. For example: production, advertising, rents, salary to human resources and so on.

6. Economic Growth and Stability:

Proper financial planning will ensure your economic growth. Gradually you will expand your wealth creation which will help you to grow financially. Important thing in someone's life is financially stability. Only way to ensure your financial stability is through economic growth and only option to ensure the same is through financial management.

7. Improve Standard of Living:

Once you have learned and taken good knowledge on financial management, this will not only provide you financial stability and peace of mind but also it will improve your standard of living. Your economic growth will transform into better standard of living.

8. Valuation of a Company:

Importance of financial management in an organization in the area of enlarging the variety of speculators and the business concern. Extreme point concerning of any business is that they will achieve maximum gain with greater efficiency. It may be related to increasing production or expanding business to other countries. A great management and financial specialists can assist in improving valuation of any company.

9. Tax Planning:

Your financial planning should also include your tax planning. When failing to plan your taxes appropriately, it will lead you spend more out of your pocket. For example: If you can analyze that current fiscal year you will be spending less

on taxes but in next year you are more likely to pay heavy taxes then you should manage your budget and saving accordingly. This will help you towards economic growth else you may run out of cash and may lead in disturbance in your investment decisions.

10. Capital Reserves:

Money have always been imaginable and possible really when the business earning rises to higher levels and expansion arises. Here is an importance of financial management in success of business by ways of expanding as well as creating capital reserves in the book of company's accounts.

Conclusion:

Think of importance of financial management points as mentioned above will benefit you in many ways. It will provide you financial stability, improve your standard of living, give you peace of mind and keep you financially stress free. It is highly recommended that you should start thinking of financial management when you can see how many benefits you get from it.

THE PRINCIPLES OF CORPORATE GOVERNANCE

Corporate governance involves a set of relationships between a company's management, its board, its shareholders and other stakeholders.

The [Cadbury Report](#) which was released in the UK in 1991 outlined that "Corporate governance is the system by which businesses are directed and controlled." Good corporate governance is a key factor in underpinning the integrity and efficiency of a company. Poor corporate governance can weaken a company's potential, can lead to financial difficulties and in some cases can cause long-term damage to a company's reputation.

A company which applies the core principles of good corporate governance; fairness, accountability, responsibility and transparency, will usually outperform other companies and will be able to attract investors, whose support can help to finance further growth.

This blog will briefly outline the role of each principle.

Fairness

Fairness refers to equal treatment, for example, all shareholders should receive equal consideration for whatever shareholdings they hold. In the UK this is protected by the [Companies Act 2006 \(CA 06\)](#). However, some companies prefer to have a [shareholder agreement](#), which can include more extensive and effective minority protection.

In addition to shareholders, there should also be fairness in the treatment of all stakeholders including employees, communities and public officials. The fairer the entity appears to stakeholders, the more likely it is that it can survive the pressure of interested parties.

Accountability

Corporate accountability refers to the obligation and responsibility to give an explanation or reason for the company's actions and conduct.

In brief:

- The board should present a balanced and understandable assessment of the company's position and prospects;
- The board is responsible for determining the nature and extent of the significant risks it is willing to take;
- The board should maintain sound risk management and internal control systems;
- The board should establish formal and transparent arrangements for corporate reporting and risk management and for maintaining an appropriate relationship with the company's auditor, and
- The board should communicate with stakeholders at regular intervals, a fair, balanced and understandable assessment of how the company is achieving its business purpose.

Responsibility

The Board of Directors are given authority to act on behalf of the company. They should therefore accept full responsibility for the powers that it is given and the authority that it exercises. The Board of Directors are responsible for overseeing the management of the business, affairs of the company, appointing the chief executive

and monitoring the performance of the company. In doing so, it is required to act in the best interests of the company.

Accountability goes hand in hand with responsibility. The Board of Directors should be made accountable to the shareholders for the way in which the company has carried out its responsibilities.

Transparency

A principle of good governance is that stakeholders should be informed about the company's activities, what it plans to do in the future and any risks involved in its business strategies.

Transparency means openness, a willingness by the company to provide clear information to shareholders and other stakeholders. For example, transparency refers to the openness and willingness to disclose financial performance figures which are truthful and accurate.

Disclosure of material matters concerning the organisation's performance and activities should be timely and accurate to ensure that all investors have access to clear, factual information which accurately reflects the financial, social and environmental position of the organisation. Organisations should clarify and make publicly known the roles and responsibilities of the board and management to provide shareholders with a level of accountability.

Transparency ensures that stakeholders can have confidence in the decision-making and management processes of a company.

Benefits of Corporate Governance

Strong corporate governance maintains investors' confidence, whose support can help to finance further growth. Companies who implement the principles of good corporate governance into working environment life will ensure corporate success and economic growth. They are the basis on which companies can grow.

It helps policy makers evaluate and improve the legal, regulatory and institutional frame work for corporate governance with a view to support economic efficiency, sustainable growth and financial stability.

Corporate Strategic goals

Strategic goals are the specific financial and non-financial objectives and results a company aims to achieve over a specific period of time, usually the next three to five years.

Strategic goals are one of three things a company's management team must articulate as part of its [strategic planning process](#), with the others being its key success factors (the important elements required to achieve its goals) and its strategic scope (the products and services that will be offered, to whom and where).

Strategic goals are important because they:

- Drive priority setting, resource allocation, capability requirements and budgeting activities
- Inform individual and team objectives used to focus and align the efforts of all employees
- Inform the marketing, operations, IT and human resources plans for the coming years
- Provide benchmarks for comparing planned and actual results

Companies use the insights from their [SWOT analysis](#) to set their goals. They are refreshed annually as part of the strategic planning process.

More about strategic goals

Strategic goals may be either “hard” or “soft”—that is precisely measurable or not. Both types of goals should be defined for a variety of stakeholders, including customers, employees, communities and shareholders. The following chart shows some examples of hard and soft goals for each stakeholder:

FUNCTIONS OF TREASURY MANAGER

Basic Function: The treasurer position is responsible for corporate [liquidity](#), [investments](#), and [risk management](#) related to the company's financial activities.

Principal Accountabilities:

1. Forecast [cash flow](#) positions, related borrowing needs, and funds available for investment

2. Ensure that sufficient funds are available to meet ongoing operational and capital investment requirements
3. Use [hedging](#) to mitigate [financial risks](#) related to the [interest rates](#) on the company's borrowings, as well as on its [foreign exchange](#) positions
4. Maintain banking relationships
5. Maintain [credit rating agency](#) relationships
6. Arrange for [equity financing](#) and [debt financing](#)
7. Invest funds
8. Monitor the activities of third parties handling outsourced treasury functions on behalf of the company
9. Advise management on the liquidity aspects of its short- and long-range planning
10. Oversee the extension of [credit](#) to customers
11. Maintain a system of policies and procedures that impose an adequate level of control over treasury activities
12. Should have a thorough understanding of [derivatives](#), hedging, investments, bank account management, and international funds flows.

Supervises: Treasury staff

SOURCES OF FUNDS

[Long-Term Sources of Finance](#)

[Long-term financing](#) means capital requirements for a period of more than 5 years to 10, 15, 20 years or maybe more depending on other factors. Capital expenditures in fixed assets like plant and machinery, land and building, etc of business are funded using long-term sources of finance. Part of working capital which permanently stays with the business is also financed with long-term sources of funds. Long-term financing sources can be in the form of any of them:

- [Share Capital or Equity Shares](#)
- [Preference Capital or Preference Shares](#)
- Retained Earnings or Internal Accruals
- [Debenture](#) / [Bonds](#)
- [Term Loans](#) from Financial Institutes, Government, and Commercial Banks
- [Venture Funding](#)
- Asset Securitization
- International Financing by way of [Euro Issue](#), Foreign Currency Loans, [ADR](#), [GDR](#), etc.

Medium Term Sources of Finance

Medium term financing means financing for a period of 3 to 5 years and is used generally for two reasons. One, when long-term capital is not available for the time being and second when deferred revenue expenditures like advertisements are made which are to be written off over a period of 3 to 5 years. Medium term financing sources can in the form of one of them:

- Preference Capital or Preference Shares
- Debenture / Bonds
- Medium Term Loans from
 - Financial Institutes
 - Government, and
 - Commercial Banks
- [Lease Finance](#)
- [Hire Purchase Finance](#)

Short Term Sources of Finance

Short term financing means financing for a period of less than 1 year. The need for short-term finance arises to finance the current assets of a business like an inventory of raw material and finished goods, debtors, minimum cash and bank balance etc.

Short-term financing is also named as working capital financing. Short term finances are available in the form of:

- [Trade Credit](#)
- Short Term Loans like [Working Capital Loans](#) from Commercial Banks
- Fixed Deposits for a period of 1 year or less
- Advances received from customers
- Creditors
- Payables
- [Factoring](#) Services
- [Bill Discounting](#) etc.

CAPITAL BUDGETING

Capital budgeting is a planning process for appraising long-term investments. As such, it is otherwise known as investment appraisal. It is an evaluation for a major capital or investment used in determining whether projects such as building a new plant or investing in a long-term venture in an organization is to be embarked upon or not. Mostly, the cash flows (cash inflow and cash outflow) of the project are used to evaluate the project to know whether it is viable or not and whether the set target would be achieved or not.

Capital budgeting can be explained in the context of a firm's decision to invest its current funds in long-term activities in anticipation of an expected flow of future benefits over a number of years. However, the investment decisions could be in form of:

- a. Construction of new factory
- b. Procurement of new vehicle
- c. Acquisition of plant and machinery
- d. Research and development projects
- e. Replacement of machinery
- f. Introduction of new product etc.

The financial manager should give due consideration to the following factors when capital budgeting decisions are involved:

- i. Availability of investment capital and its alternative uses
- ii. The huge expenditure cash outlays
- iii. The gestation period between initial expenditure and returns, and
- iv. The expectation of higher returns because of factors (i) and (ii) above.

Characteristics of capital Budgeting

Capital budgeting (Capital expenditure) unlike day-to-day expenditure or short-term investment has the following features: -

1. They involve relatively long period between the initial outlay and the anticipated recoupment of the capital.
2. They involve large sum of money or outlay.
3. They are very risky as they involve future cash flows that are subject to risk and uncertainty.
4. They involve potentially large anticipated benefits.
5. Their benefits usually spread over a long period of time usually two or more years.
6. They involve irreversible decision as they involve large outlay and long life.

Because of the aforementioned features, it is therefore important that adequate procedure must be in place to ensure effective utilization of firm's resources. This would help in safeguarding the assets and resources of the business.

OBJECTIVES OF INVESTMENT EVALUATION

- a. To ensure that the limited capital resources available are distributed to wealth creating capital projects which make the best contribution to the achievement of corporate goals.
- b. To ensure that poor or ill-defined proposals are rejected or further refined.
- c. To evaluate various investment proposals and select the most viable ones.
- d. To avoid under and over investment.

PROCEDURE INVOLVE IN CAPITAL BUDGETING DECISION

One of the arrears of decision making of a firm, which require serious attention is the capital investment decision. This has to do with the crucial influence of it on the profitability of the firm as well as its importance in achieving the goals of the firm. As such, a firm, when making capital budgeting or investment decision, should adhere to the following;

- Identify the corporate objective of the organization
- Identify the possible projects (investment profile/ portfolio).
- Determination of budget estimate.
- Search for and development of the projects.
- Evaluation of alternative investment proposals
- Authorization of selected projects by the management.
- Implementation, Monitoring and Control.

PRINCIPLES OF CAPITAL EXPENDITURE AND CONTROL

To ensure efficient allocation of scarce resources, each capital project should pass through the following process: -

Justification: Each project should be justified by a statement of the reasons for the expenditure and a forecast of the profitability.

Authorization: Before any capital expenditure is incurred or entered into, it should be authorized within the overall capital budget, which incorporates all projects in progress or to commence during the budget period.

Authentication: As a project being carried out, all expenditures should be authenticated as being within the previously authorized budget for the project.

Control: Each project should have some means of identification and cost collection e.g. a code number, which serves as identification number that ensures all expenditures can be collected and compared with the budget stage by stage.

Monitoring: details of completion compared with estimates should be published for guidance in future estimating.

CAPITAL BUDGETING TECHNIQUES

There are two major methods of appraising capital projects; these are: -

- a. The Non-discounted cash flow (Traditional) method
- b. The discounted cash flow (Modern) method

TRADITIONAL METHOD

These are methods or techniques of appraising investments, which do not take into consideration, the time value of money in evaluating investment proposals or projects.

They include:

- Payback Period (PBP) method
- Accounting Rate of Return (ARR) method

PAYBACK PERIOD (PBP) METHOD:

This technique is used to determine how long an investment will be recouped. As such, it is used to determine how quickly a project or investment repays its outlay. In other words, payback period is the number of years it takes a project to pay the amount invested in the project. In calculating Payback Period, the incremental cash inflow associated with the project should be used. The incremental cash inflow unless otherwise stated should be taken as the profit generated by the project but before charging depreciation, absorption of overheads and other notional accounting charges.

Where cash flows from an investment or project are regular or same, the Payback Period is calculated using this formula:

Initial outlay divides by Annual net cash flow.

For example: if a project costs ₦80,000 and constant annual cash flow of ₦20,000 p.a. for a period of 6 years. Then, Payback Period is determined by as:

$$= \frac{\text{₦80,000}}{\text{₦20,000}} = 4 \text{ years.}$$

It means the project would pay back the money invested at the end of the fourth year.

For projects with irregular or not similar cash inflows. The Payback Period is the time it will take the cumulative cash inflow from the project to adequate the initial investment outlay of the project. This is determined using the tabular format as follows:

Example 1: A three-year project with initial investment of ₦30,000 and annual cash inflow as follows: year 1 – ₦15,000, year 2 – ₦12,000, and year 3 – ₦13,000.

Example 2: BAM Ltd is a manufacturing outfit, having a project which involves an immediate cash outlay of ₦200,000. The company estimates that the net cash inflows from the project will be as follows:

Year	Cash inflows (₦)
1	20,000
2	40,000
3	220,000
4	80,000

Calculate the payback period for the above project.

DECISION RULE:

Independent project: A project is accepted when the payback period is less than or equal to that set by the management. They should be rejected if the payback period is more than the management benchmark.

Mutually exclusives projects: These are projects serve the same purpose and compete with each other for the organization's resources. In this case, the organization cannot go for the two projects. If the organization selects or accepts one of the projects, the other one has to be rejected because they all serve the same purpose. In this situation, the project with least payback period will be accepted.

DISCOUNTED PAYBACK PERIOD METHOD

The principles and decision rules are the same as in the normal payback method. The only difference is that, the cash flows to be used are discounted at the given or appropriate cost of capital. Therefore, this version of payback technique, does not suffer from the disadvantage of ignoring time value of money.

Illustration:

Madam Esther is seeking a bank to grant her a loan of ₦400,000, which she is to invest in a poultry project. She estimates that the project will yield the following returns annually for the next five years.

Year	1	2	3	4	5
₦	120,000	120,000	160,000	120,000	80,000

Required: calculate the discounted payback of the project, if the cost of capital is 10% per annum.

ADVANTAGES OF PAYBACK PERIOD

1. It is simple to calculate and understand
2. The fact that it focuses on the shortness of the project's life span, makes it to be the least affected by uncertainties
3. Since payback period emphasis early repayment of amount invested. It can assist in liquidity problem
4. If it is agreed that risk is a function of time; and the longer it takes to get the outlay back, the higher the level of risk. Then payback period method is often used in risk analysis and assists in risk management
5. Payback period uses cash profit or incremental cash flow, which is superior to Accounting profit
6. In screening new products, Payback period is used as a first or initial screening process.

DISADVANTAGES OF PAYBACK PERIOD

1. It does not into consideration the times value of money
2. It does not also consider the cash flows occurring after the pay-back period
3. Because there are no rules for setting the maximum acceptable payback period by management, it is usually subjective

4. It has a problem of definition of what the outlay of a project is, which is the starting point of the payback period.

ACCOUNTING RATE OF RETURN (ARR):

This is accounting based technique of appraising capital projects, which is based on return on capital employed. This is also referred to as Return on Investment Method. This technique uses Accounting profit and the capital employed instead of cash flows or economic profit. A capital investment project may be appraised by computing the Accounting Rate of Return (ARR) and comparing it with cut-off or huddles rate set by the management. There are several ways by which ARR can be calculated: -

- a.
$$\text{ARR} = \frac{\text{Average Accounting profits after depreciation, interest but before taxation} \times 100}{\text{Initial investment}} 1$$
- b.
$$\text{ARR} = \frac{\text{Average Accounting profits after depreciation, interest but before taxation} \times 100}{\text{Average capital invested}} 1$$

Illustration:

SLT recently convinced his friends and relations to grant him a loan of N200,000, which he intends to invest in a farming project. He estimates that the project will yield the following returns annually for the next five years.

Year	N
1	60,000
2	60,000
3	80,0000
4	60,000
5	40,000

There was no scrap value at the end of the fifth year and the company desires to evaluate the project on the basis of accounting rate of return.

Required: Provide the accounting rate of return of this project on the assumption that the annual returns are the profits after depreciation but before taxation.

ADVANTAGES OF ARR

1. The ARR is easy to compute and simple to understand
2. The ARR makes use of profit generated throughout the project life. In other words, ARR differs from payback period and it takes into consideration all profits of the project
3. The ARR uses accounting information or data
4. It used to compare two or more companies' in terms of their performance.

DISADVANTAGES OF ARR

1. The ARR being one of the traditional methods or techniques does not take into consideration the time value of money.
2. It does not use cash profit as a measure of profit. That is, it makes use of accounting profit rather than incremental cash flow.
3. The method lacks rules for setting of minimum acceptable ARR by the management. As such, the setting of minimum Accounting Rate of Return (ARR) by management is usually subjective.
4. This method pays no attention to risk. It also does not take into consideration the attitude of the management towards risk.
5. For mutual exclusive projects, this method goes for or favours projects with higher ARR not considering the cash involve. For example, a project with ARR of 10% having investment of ₦500,000 and another with ARR of 15% having investment of ₦100,000.

DISCOUNTING CASH FLOW TECHNIQUES

Discounted cash flow techniques otherwise known as the Modern methods take into consideration of the time value of money. It discounts the cash flows to their present values using the prevailing interest rate or cost of capital in order to appraise or evaluate capital projects. As such, it takes care of the overall profitability of the projects and the timing of returns. In other words, the timing of the cash flows is taken into consideration by means of discounting.

Discounted cash flow techniques consider the cash receipt and payments made or foregone as a result of taking a course of action. Only relevant costs should be considered. A discounted cash flow technique therefore, applies the concept of incremental cost in assessing profitability or otherwise of investment projects. The following are some of the discounted cash flow techniques:

Net Present Value (NPV):

This method accepts the fact that the value attached to Naira at different times are not same. As the time increases, the value attach to Naira decreases. Therefore, this method is the sum project's cash flow over the life of the project discounted at the company's cost of capital. In calculating NPV, incremental cash flow techniques could be used with a view to maximizing shareholders' wealth as a result of undertaking a particular project. NPV depends on the discount factor, the magnitude of the outlay, the magnitude of the project cash inflows etc.

Decision Rule:

Independent Projects: Accept project with positive NPV. In other words, if a project's NPV is greater than zero, accept it. If the NPV is less than zero, reject it. If NPV is zero, the final decision will be based on qualitative factors.

Mutually Exclusive Projects: In the case of mutually exclusive projects the project with the higher positive NPV should be accepted.

Illustration:

Use the example of John above at 10% cost of capital, calculate the NPV of the project.

ADVANTAGES OF NPV:

1. The NPV method recognizes or takes into consideration, the time value of money
2. The NPV method unlike the payback period, consider all cash flows over the project's life span

3. The NPV method represents a way of determining the profitability of a project.
This reflects in the wealth of the shareholders
4. The method instead of using accounting profit it uses incremental cash flow
5. The method is consistent with the wealth maximization objective of the firm
6. The NPV method has a well-defined decision rule. That is to say, it has a clear-cut accept or reject decision.

DISADVANTAGES OF NPV

1. It is difficult to estimate the cost of capital in practice
2. NPV does not take inflation into consideration
3. It is simple to compute
4. It does not consider the likelihood of changes in the project life, cost of capital, cash flow, etc.
5. It is not appropriate in assessing unequal projects. For example, the one with higher NPV might not necessary be the most profitable when you consider the value of the investment.

INTERNAL RATE OF RETURN (IRR)

The internal rate of return otherwise known as the Discounted Cash Flow yield, the cut-off rate or hurdle rate is the discount rate that equates the NPV of a project to zero. It is the break-even cost of capital. It is the cost of capital at which there is no additional wealth to be generated out of the investment opportunity. The IRR usually represents the maximum acceptable discount rate or cost of borrowing for a project. That is a cost of borrowing or cost of capital above which the project should be rejected because it will not be viable. While for a cost of capital below the IRR, the project should be accepted because it will be profitable or viable. The IRR is usually calculated by trial and error as follows:

Step 1: Compute the NPV of the project using the company's cost of capital. If the examiner gives no cost of capital, use any reasonable rate between, for example 5%, 10% etc.

Step 2: If the NPV is positive, it means the trial rate is low; therefore, a higher rate should be used, which will produce a negative NPV. Note that we could interpolate whether the second NPV is negative or not, provided we strictly maintain the formula. But, where the first NPV is negative, you need to lower the rate and then recompute the second NPV that should be positive.

Step 3: to generate the IRR, interpolate the results obtained in (1) and (2) as follows:

$$R1 + \left\{ \frac{NPV1}{NPV1 + NPV2} \right\} \times (R2 - R1)$$

Where: R1 = lower rate/ lower cost of capital
 R2 = higher rate/ higher cost of capital
 NPV1 = the NPV obtained by using the lower cost of capital
 NPV2 = NPV obtained from the higher cost of capital

Decision rule:

Independent projects: A project should be accepted if its IRR is equal to or greater than the company's cost of capital

Mutually exclusive projects: The project with the higher IRR should be chosen. However, the IRR of the chosen project should be greater than the company's cost of capital.

Illustration:

Use the example of John above to calculate the IRR

ADVANTAGES OF IRR:

- i. Just like the NPV method, IRR method also considers the time value of money
- ii. It uses or considers the cash flows over the entire project's life span
- iii. Unlike the NPV, the company's cost of capital is not a precondition for the computation of IRR

- iv. The IRR figure is familiar, which makes it convenient for users of the data to interpreted
- v. The IRR method takes care of the problem of which cost of borrowing is appropriate to be used

DISADVANTAGES OF IRR

- i. The IRR method is difficult to understand and use in practice
- ii. When cash flows are non-conventional, IRR will produce multiple rates
- iii. It does not show the additional value in monetary terms to be generated.

ADVANTAGES OF NPV OVER IRR

- i. NPV method gives an absolute measurement of the benefit associated with a project i.e. the benefit of project is qualified in Naira and kobo. IRR on the other hand, gives a relative measure i.e. measure in percentage
- ii. Where projects with different outlays are being considered, IRR can give a misleading conclusion
- iii. NPV is additive whereas IRR is not
- iv. When there is capital rationing, NPV can be modified by calculating profitability index to rank the project. IRR cannot be so modified
- v. When there are changes in cost of capital, it can probably be incorporated into NPV calculation whereas in IRR, such changes cannot be so modified
- vi. When there are non-conventional cash flows (i.e. cash flow involving negative and positive), IRR suffers the problem of multiple rates.

CAPITAL RATIONING

situation is one in which a company does not have sufficient fund to execute worthwhile investment project. Under this situation, accompany has projects with NPV whose combined outlays exceed all available finance to the company for the same period.

Capital Rationing is the technique for selecting projects during a period of funds restriction which normally requires the ranking of these projects in a descending order

of desirability and accepting them in that order until all available funds have been exhausted.

Single Period Capital Rationing: where restriction is for only one period, Probability Index is used.

Probability Index: This concept is based on the contribution per limiting factor approach. It is actually a benefit/ cost analysis of projects, that is, the naira receivable per Naira of expenditure. It can be measured as the ratio of NPV or in some cases, Gross Present Value (GPV) of a project to the outlay required for the project during the year of restriction. The PV's are the contributions while the expenditure or the outlays are the limiting factors. However, two definitions of PI are allowable as follows: -

$$PI = \frac{\text{NPV of a project}}{\text{Outlay required during the year of restriction}}$$

$$PI = \frac{\text{Gross Present Value (GPV) of a project}}{\text{Outlay required during the year of restriction}}$$

Where: GPV will normally equal to NPV + initial Outlay

Steps to be Taken in a Single Period Capital Rationing Situation

The following steps are recommended when questions are being attempted concerning a single period capital rationing situation:

- a. Identify the year of restriction;
- b. Calculate the NPV of projects (if not given);
- c. Rank all projects using PI;
- d. Allocate available finance or funds to all projects in a descending order of the PI;

- e. If a project does not require outlay during the year of restriction, its PI would be an infinite sum, (that is, $NPV + 0$) and such projects must be ranked first and must be selected.

Illustration:

Bukky Aina Ltd has a capital budget of 500,000 for the year to 30th June, 2009, the available projects have been identified and quantified by the Technical Director and the Works Manager as listed below. The individual project's related profitability index has been computed by a Finance Manager team and stated hereunder.

Projects	Initial Outlay	Profitability Index
	₦	₦
A	250,000	1.10
B	100,000	0.95
C	200,000	1.25
D	200,000	1.23
E	250,000	1.05
F	100,000	1.20
G	50,000	0.99

Required: - a. Which projects should the company invest in?

b. What difference would the absence of capital rationing make to your selection in (a) above?

Different situations of capital Rationing

The different situations of capital rationing are as follows:

- Where projects are divisible
- Where projects are not divisible
- Where projects are mutually dependent
- Where projects are mutually exclusive

Divisible Projects:

In this case, there is implicit linearity assumption between the initial outlays and the NPVs of projects, this follows from the basic assumption that fractions of a project can be undertaken. Therefore, a fraction investment in the outlay would yield a proportionate fraction return in NPV e.g. investment of 20% in outlay, would yield 20% of NPV.

Indivisible Projects:

These are projects that cannot be undertaken in fractions. In this case, there may be surplus funds which represent the balance of available funds after allocation that cannot meet the requirement in full for the remaining projects. In this situation, there is a need to invest such surplus funds. This is one of such situations where a company can invest at a rate below its cost of capital.

Illustration:

New Global Ltd is experiencing a shortage of funds for investment in the current year, when only ₦100,000 is available for investment. No fund shortages are foreseen thereafter. The cost of investment funds is 20%. The following projects are available:

Projects	1	2	3	4	5	6
Initial outlay(₦)	50,000	80,000	60,000	30,000	25,000	40,000
Annual receipts						
From project to						
Perpetuity(₦)	15,000	20,000	18,000	10,000	8,000	10,000

You are required to advise Management on the projects which you would recommend for acceptance if they were.

- a. Divisible, b. indivisible

Would your answer to (b) change if surplus funds could be invested at 12% (ignore taxation).

Where projects are mutually dependent:

In this case, an acceptance of the mutually dependent projects automatically implies the acceptance of the remaining mutually projects. For example, if projects A and B are mutually dependent, it means that one can only accept or reject both projects and

not undertake only one of them. However, any of the two methods may be adopted as follows:

- a. In ranking all projects, one can decide to rank all the mutually dependent projects as soon as or immediately after any of the mutually dependent projects is ranked irrespective of the sizes of their respective profitability index. For instance, if projects A and B are mutually dependent in a group of the under listed projects, the ranking can be done as follows:

Projects	PI	Initial Outlay	Ranking
A	1.1	200	3 rd
B	1.6	100	2 nd
C	1.7	200	1 st
D	1.2	300	5 th
E	1.3	400	4 th

Therefore, one can select and allocate funds in that order as above.

- b. If one wants to arrive at a more accurate solution, he has to combine all mutually dependent projects into a unique single project. Where the latter would have as its initial outlay, a combined total of all initial outlays of the mutually dependent projects. The PI would be a weighted average PI, where the weight to be attached to the PI will be initial outlays of the prospective projects. Using the same group as above, the projects can be ranked after a combination of projects A and B as follows:

PI x Initial outlay

300

$$PI = \frac{(1.1 \times 200) + (1.6 \times 100)}{300}$$

$$= 1.27$$

A and B = 1.27

The new ranking of project is

1 st	2 nd	3 rd	4 th
C = 1.7	E = 1.3	AB = 1.27	D = 1.2

Where projects are Mutually Exclusive:

In this case, an acceptance of one project's group implies the rejection of all other project groups to resolve this issue, there is need to modify the ranking procedures and it is done as follows:

- i. Create as many groups of projects as long they are mutually exclusive i.e. each group must contain only one of the mutually exclusive projects and would of course exclude the others.
- ii. Rank and select projects in each group
- iii. Calculate the total NPV's of the projects in each group
- iv. The decision would be to accept projects from the group that produce maximum total NPVs.

Illustration:

Abubakar, Dele and Edwin have just received their gratuities which amounted to ₦250,000 and they are prepared to invest in a new venture XYZ Ltd the bank of Nigeria Ltd has expressed the desire to grant them long-term loan of up to ₦950,000. They have presented the following investment proposals to you for financial advice:

Projects	1	2	3	4	5	6	7	8	9	10
PI	1.19	1.20	1.17	1.20	1.22	1.15	1.10	1.15	1.22	1.16
Outlays ('000)	150	150	200	300	400	400	300	200	100	100

They expected cost of capital is 15%. Projects 3 and 6 are mutually exclusive while projects 1 and 4 are mutually dependent.

Required:

- a. As a financial adviser, what projects would you recommend that the new venture should embark upon?
- b. What is the opportunity cost of accepted project?

Multi-Period Capital Rationing

Where capital is restricted in more than one period, the requirement is likely in the least, a formulation of the linear programming required, to select projects which will maximize the NPVs for the company. For this reason, the following steps will be required:

- a. Identify the relevant variables (e.g. available funds, required fund for each project, years of restriction of respective projects and the resultant NPVs etc.
- b. Specify these variables in a form adaptable to mathematical manipulation by representing the variables in (a) above with symbols let “X” represent the MICHKA project, etc. or let “Y” represent the fraction of MAIHA project accepted.
- c. State the objective function, which normally is the maximization of the NPVs of all the projects.
- d. Specify the constraints as follows:
 - i. **Financial constraint:** This ensures that total required funds for all the projects and other operations of the company do not exceed the available funds for each corresponding period.
 - ii. **Non-negativity constraint:** This ensures that projects are either accepted or rejected and in no situation would a negative portion of a project be accepted.
 - iii. **Logical constraint:** This ensures that fractions of projects can be undertaken whereas no projects will be repeated.

Note: Steps (b), (c) and (d) will constitute the linear programming formula or plan for multi-period capital rationing.

Limitation of Capital Rationing: -

The following constitute the limitations of capital rationing:

- a. The assumption of divisibility of projects may not be possible in practice for all projects.
- b. In addition to (a), we also assume linearity between outlays and NPVs of projects, for example, a fractional investment in outlay will yield the same fraction of NPV. In practice, because of economies and scale, this may not hold.
- c. On many occasions, capital rationing treats projects in isolation. It does not recognize the interdependence of projects, for example, an investment in a

project may result in substantial savings only if another project is undertaken, which capital rationing ignores.

- d. Where more than two projects are involved and capital is restricted in more than one period, the profitability index approach will no longer be sufficient for resolving project acceptance under capital rationing. In such cases, linear programming technique may be applied.

THE THEORY OF CAPITAL STRUCTURE & DIVIDEND POLICY DECISION